Command Prompt - nslookup

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Microsoft Windows [Version 10.0.19045.6466]
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C:\Users\TCS>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a             Displays current ARP entries by interrogating the current
                 protocol data.  If inet_addr is specified, the IP and Physical
                 addresses for only the specified computer are displayed.  If
                 more than one network interface uses ARP, entries for each ARP
                 table are displayed.
  -g             Same as -a.
  -v             Displays current ARP entries in verbose mode.  All invalid
                 entries and entries on the loop-back interface will be shown.
  inet_addr      Specifies an internet address.
  -N if_addr     Displays the ARP entries for the network interface specified
                 by if_addr.
  -d             Deletes the host specified by inet_addr. inet_addr may be
                 wildcarded with * to delete all hosts.
  -s             Adds the host and associates the Internet address inet_addr
                 with the Physical address eth_addr.  The Physical address is
                 given as 6 hexadecimal bytes separated by hyphens. The entry
                 is permanent.
  eth_addr       Specifies a physical address.
  if_addr        If present, this specifies the Internet address of the
                 interface whose address translation table should be modified.
                 If not present, the first applicable interface will be used.
Example:
  > arp -s 157.55.85.212   00-aa-00-62-c6-09  .... Adds a static entry.
  > arp -a                                    .... Displays the arp table.

C:\Users\TCS>hostname
DESKTOP-F4RM5ER
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C:\Users\TCS>hostname
DESKTOP-F4RM5ER

C:\Users\TCS>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::d51f:7aad:f478:5629%7
   IPv4 Address. . . . . . . . . . . : 172.16.11.192
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Default Gateway . . . . . . . . . : 172.16.8.1

C:\Users\TCS>nbtstat

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.
```

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C:\Users\TCS>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    172.16.11.192:51810    4.213.25.240:https     ESTABLISHED
  TCP    172.16.11.192:51812    a23-202-229-23:https   CLOSE_WAIT
  TCP    172.16.11.192:51813    a23-202-229-23:https   CLOSE_WAIT
  TCP    172.16.11.192:51814    a23-211-60-142:https   CLOSE_WAIT
  TCP    172.16.11.192:51815    a23-211-60-142:https   CLOSE_WAIT
  TCP    172.16.11.192:51816    a23-211-60-142:https   CLOSE_WAIT
  TCP    172.16.11.192:51817    a23-202-229-23:https   CLOSE_WAIT
  TCP    172.16.11.192:51818    a23-202-229-23:https   CLOSE_WAIT
  TCP    172.16.11.192:51819    a23-202-229-23:https   CLOSE_WAIT
  TCP    172.16.11.192:51820    a23-199-67-57:https    CLOSE_WAIT
  TCP    172.16.11.192:51821    a23-199-67-57:https    CLOSE_WAIT
  TCP    172.16.11.192:52699    123:http               TIME_WAIT
  TCP    172.16.11.192:52700    123:http               TIME_WAIT
  TCP    172.16.11.192:52701    150.171.28.12:https    ESTABLISHED
  TCP    172.16.11.192:52702    a23-202-229-22:https   CLOSE_WAIT
  TCP    172.16.11.192:52703    4.207.44.66:https      TIME_WAIT
  TCP    172.16.11.192:52704    a23-202-229-22:https   ESTABLISHED
  TCP    172.16.11.192:52705    4.213.25.242:https     ESTABLISHED
  TCP    172.16.11.192:52706    123:http               ESTABLISHED

C:\Users\TCS>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list      Loose source route along host-list.
    -h maximum_hops   Maximum number of hops to search for target.
    -i address        Use the specified source address.
    -n                Do not resolve addresses to hostnames.
    -p period         Wait period milliseconds between pings.
    -q num_queries    Number of queries per hop.
    -w timeout        Wait timeout milliseconds for each reply.
    -4                Force using IPv4.
    -6                Force using IPv6.
```

```
C:\Users\TCS>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t              Ping the specified host until stopped.
                    To see statistics and continue - type Control-Break;
                    To stop - type Control-C.
    -a              Resolve addresses to hostnames.
    -n count        Number of echo requests to send.
    -l size         Send buffer size.
    -f              Set Don't Fragment flag in packet (IPv4-only).
    -i TTL          Time To Live.
    -v TOS          Type Of Service (IPv4-only. This setting has been deprecated
                    and has no effect on the type of service field in the IP
                    Header).
    -r count        Record route for count hops (IPv4-only).
    -s count        Timestamp for count hops (IPv4-only).
    -j host-list    Loose source route along host-list (IPv4-only).
    -k host-list    Strict source route along host-list (IPv4-only).
    -w timeout      Timeout in milliseconds to wait for each reply.
    -R              Use routing header to test reverse route also (IPv6-only).
                    Per RFC 5095 the use of this routing header has been
                    deprecated. Some systems may drop echo requests if
                    this header is used.
    -S srcaddr      Source address to use.
    -c compartment  Routing compartment identifier.
    -p              Ping a Hyper-V Network Virtualization provider address.
    -4              Force using IPv4.
    -6              Force using IPv6.
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```
C:\Users\TCS>route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

  -f            Clears the routing tables of all gateway entries.  If this is
                used in conjunction with one of the commands, the tables are
                cleared prior to running the command.

  -p            When used with the ADD command, makes a route persistent across
                boots of the system. By default, routes are not preserved
                when the system is restarted. Ignored for all other commands,
                which always affect the appropriate persistent routes.

  -4            Force using IPv4.

  -6            Force using IPv6.

  command       One of these:
                  PRINT     Prints  a route
                  ADD       Adds    a route
                  DELETE    Deletes a route
                  CHANGE    Modifies an existing route
  destination   Specifies the host.
  MASK          Specifies that the next parameter is the 'netmask' value.
  netmask       Specifies a subnet mask value for this route entry.
                If not specified, it defaults to 255.255.255.255.
  gateway       Specifies gateway.
  interface     the interface number for the specified route.
  METRIC        specifies the metric, ie. cost for the destination.
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All symbolic names used for destination are looked up in the network database
file NETWORKS. The symbolic names for gateway are looked up in the host name
database file HOSTS.

If the command is PRINT or DELETE. Destination or gateway can be a wildcard,
(wildcard is specified as a star '*'), or the gateway argument may be omitted.

If Dest contains a * or ?, it is treated as a shell pattern, and only
matching destination routes are printed. The '*' matches any string,
and '?' matches any one char. Examples: 157.*.1, 157.*, 127.*, *224*.

Pattern match is only allowed in PRINT command.
Diagnostic Notes:
    Invalid MASK generates an error, that is when (DEST & MASK) != DEST.
    Example> route ADD 157.0.0.0 MASK 155.0.0.0 157.55.80.1 IF 1
             The route addition failed: The specified mask parameter is invalid. (Destination & Mask) != Destination.

Examples:

    > route PRINT
    > route PRINT -4
    > route PRINT -6
    > route PRINT 157*          .... Only prints those matching 157*

    > route ADD 157.0.0.0 MASK 255.0.0.0  157.55.80.1 METRIC 3 IF 2
             destination^      ^mask      ^gateway     metric^    ^
                                                       Interface^
      If IF is not given, it tries to find the best interface for a given
      gateway.
    > route ADD 3ffe::/32 3ffe::1

    > route CHANGE 157.0.0.0 MASK 255.0.0.0 157.55.80.5 METRIC 2 IF 2

      CHANGE is used to modify gateway and/or metric only.

    > route DELETE 157.0.0.0
    > route DELETE 3ffe::/32
```

```
C:\Users\TCS>nslookup www.google.com
Server:  UnKnown
Address:  172.16.8.1

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:4827:7700::
          2001:4860:482b:7700::
          2001:4860:4828:7700::
          2001:4860:4826:7700::
          2001:4860:482d:7700::
          2001:4860:4829:7700::
          2001:4860:482c:7700::
          2001:4860:482a:7700::
          142.251.156.119
          142.251.153.119
          142.251.157.119
          142.251.152.119
          142.251.151.119
          142.251.154.119
          142.251.155.119
          142.251.150.119


C:\Users\TCS>
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